# R and clinical/lab data

2026 edition

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# Welcome

Welcome to this course, here you can find the notebooks used in the PhD course of 2026.

# Chapter 1

# Dataframes, vectors and numbers

## 1.1 Dataframes

The usual data you will deal with are tabular (rows x columns) data, usually improperly called "Excel files".

The standard formats for raw data files in data analyses are *.csv*, *.tsv* and *.txt* files; these should be our starting point.

## 1.2 Import file

The first thing to do is to import the file into R session. We can do it with several functions, but the standard one for *.csv* files is `read.csv`.

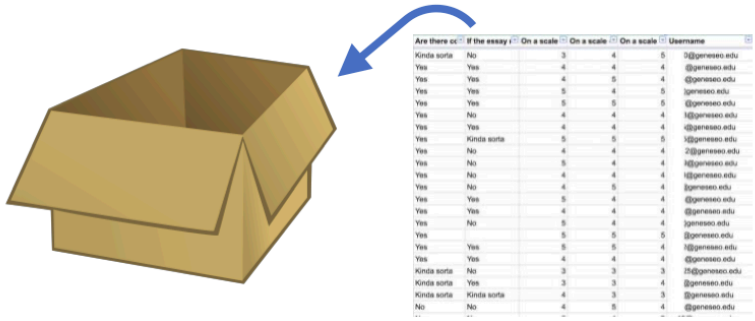
Let's see how to do so:

```
my_data <- read.csv("data/input_df.csv")
```

#### 1.2.0.1 Notes about variables

Variables are the "memory" of an R session, they represent box where R stores data/values you want.

You can think them as boxes that have a name on them:



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| Kinda sorta | Kinda sorta | 4 | 3 | 3 | )geneseo.edu |
| No | No | 4 | 5 | 4 | @geneseo.edu |

There are important **rules** about variables:

- In each variable you can store one single item (a number, a string, a dataframe, a list, ...)
- variable names can start only with a letter (uppercase or lowercase)
- variable names can contain **only** letters, numbers, underscores (_) and dots (.). **No spaces, no dash (-) or any other special characters**
- there could be only ONE variable with a specific name, saving an object with the same of another results in the substitution of the first one
- suggestion: variable names should be meaningful to you (and to the ones you share the code with): e.g. `x` is not a good name, while `patients_glucose_data` is preferred

### 1.2.1 First steps on a dataframe

When importing a dataframe, there is a fundamental step to do: *inspection*. Inspection helps us to understand how the dataframe is structured, which type

of data we have, which columns etc.

To do it, we use 3 functions that you need to remember (tatoo yourself): `summary`, `str` and `head`. Let's see them:

```
summary(my_data)
```

```
     sample        timepoint           genotype          treatment
 Min.   :  1.00   Length:192         Length:192         Length:192
 1st Qu.: 48.75   Class :character   Class :character   Class :character
 Median : 96.50   Mode  :character   Mode  :character   Mode  :character
 Mean   : 96.50
 3rd Qu.:144.25
 Max.   :192.00
      area        intensity
 Min.   :3000   Min.   : 9130286
 1st Qu.:3259   1st Qu.:16283904
 Median :3517   Median :18894104
 Mean   :3508   Mean   :19419175
 3rd Qu.:3769   3rd Qu.:22090201
 Max.   :3988   Max.   :29812114
```

```
str(my_data)
```

```
'data.frame':   192 obs. of  6 variables:
 $ sample   : int  1 2 3 4 5 6 7 8 9 10 ...
 $ timepoint: chr  "P1" "P1" "P1" "P1" ...
 $ genotype : chr  "wt" "wt" "wt" "wt" ...
 $ treatment: chr  "n" "n" "n" "n" ...
 $ area     : num  3125 3500 3773 3538 3936 ...
 $ intensity: num  16015465 19449815 21118210 18100075 20718554 ...
```

We can see how many rows and columns we have, as well as which columns, the data type of each and the first entries.

Lastly, `head` is the function that shows us the first 6 entries of the dataframe, as a table:

```
head(my_data)
```

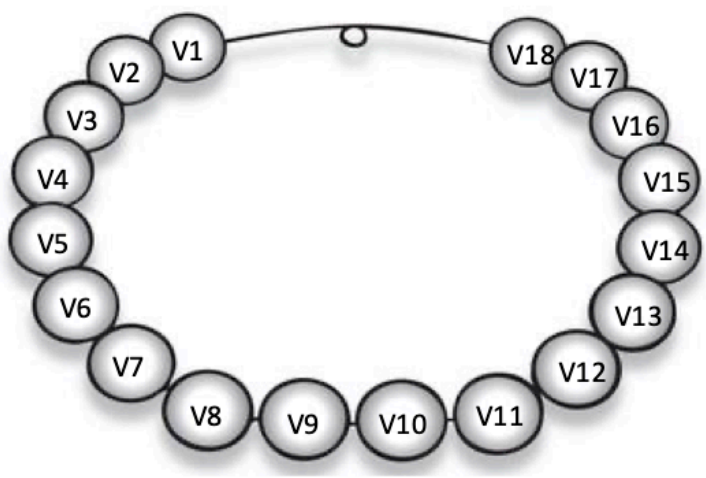
```
  sample timepoint genotype treatment     area intensity
1      1        P1       wt         n 3124.709  16015465
2      2        P1       wt         n 3499.559  19449815
3      3        P1       wt         n 3773.000  21118210
4      4        P1       wt         n 3538.320  18100075
```

```
5       5        P1       wt         n 3936.014  20718554
6       6        P1       wt         n 3646.389  19507110
```

## 1.3 Vectors

Dataframes are organized and read by columns. Each column is a **vector** in R.

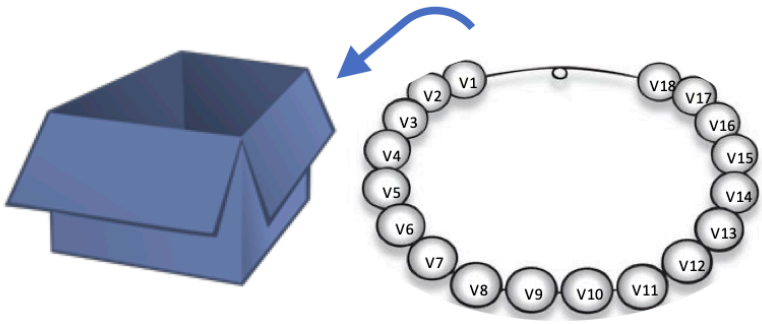
A vector is a collection of data of the **same type**; you can think of a vector as a pearl necklace, with each pearl being a single value:

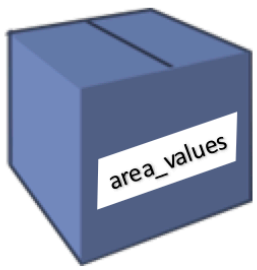



### 1.3.1 Accessing dataframe columns

For exercise purposes, we can create a new variable storing the values of the area column of the dataframe. To extract a single column from a dataframe, we use the structure `dataframe_name$column_name`.

```
area_values <- my_data$area
```




### 1.3.2 Accessing values of a vector: slicing

**Slicing** is the operation of extract values out of a variable (in this case a vector). It is done to answer these questions:

- *Which is the first value of the vector?*
- *Which is the value at position n of a vector?*
- *Which are the values from position n to m?*

There are different ways in which we can slice a vector, and today we will see indexing (through positions).

The syntax is:

- `variable_name[position]` to get one single value
- `variable_name[start:end]` to get elements from `start` to `end`

**REMEMBER**: In R, the first element is at index 1.

```
# extract one single value

first_area_value <- area_values[1]
print(first_area_value)
```

```
[1] 3124.709
```

```
# extract range of values
from_fifth_to_ninth_areas <- area_values[5:9]
print(from_fifth_to_ninth_areas)
```

```
[1] 3936.014 3646.389 3073.109 3717.063 3887.006
```

Q: *What 5:9 does in R?* A: It creates a vector containing numbers from 5 to 9, that's why it works:

```
print(5:9)
```

```
[1] 5 6 7 8 9
```

Q: *And to get last element?*

A: We need to know the lenght of the vector and use it as index. To get the length of a vector we use the function `length`:

```
length_area <- length(area_values)
print(length_area)
```

```
[1] 192
```

```
last_area <- area_values[length_area]
print(last_area)
```

```
[1] 3808.121
```

**Exercise 1.1.** Which is the 5th value of the intensity column of the dataframe?

Solution

We can do it in many steps:

```
intensity_values <- my_data$intensity

fifth_intensity <- intensity_values[5]

print(fifth_intensity)
```

```
[1] 20718554
```

Or in just one step:

```
fifth_intensity <- my_data$intensity[5]

print(fifth_intensity)
```

```
[1] 20718554
```

#### 1.3.2.1 Replace a value by index

We can replace of a vector using indexes.

**Note**: new values should be of the same type (numbers for numbers, strings for strings ecc), otherwise everything will be changed to strings (most of the time).

To change a single value, we can type `variable_name[position] <- new_value`; to change multiple values `variable_name[start:end] <- new_values`.

---

### 1.3.3 Numbers in R

Now that we know how to get and extract numbers from numeric vectors, let's see what we can do with numbers in R.

#### 1.3.3.1 Declare a new numeric variable

To declare a new numeric variable, we use the statement `variable_name <- number`:

```
conversion_factor <- 1000

print(conversion_factor)
```

```
[1] 1000
```

```r
print(typeof(conversion_factor))
```

```
[1] "double"
```

Decimal point numbers wants the dot `.` as decimal separator:

```r
decimal_n <- 215.6
print(decimal_n)
```

```
[1] 215.6
```

```r
print(typeof(decimal_n))
```

```
[1] "double"
```

#### 1.3.3.2 Arithmetical operations

We can perform all sort of matematical operation:

```r
# sum

print(2 + 5)
```

```
[1] 7
```

```r
# subtraction

print(7 - 2.5)
```

```
[1] 4.5
```

```r
# multiplication
print(12 * 5)
```

```
[1] 60
```

```r
# division
print(105 / 43)
```

```
[1] 2.44186
```

```r
# exponential
print(5 ** 2)
```

```
[1] 25
```

```r
print(5 ^ 3)
```

```
[1] 125
```

```r
# square root
print(sqrt(12))
```

```
[1] 3.464102
```

We can combine them all to do more complex operations. For example, you can resolve this equation $x^2 - 7x + 12 = 0$. We know that the formula to resolve this equation is: $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$, so we can reconstruct them in R:

```r
a <- 1
b <- -7
c <- 12

x1 <- (-b + sqrt(b**2 - 4*a*c)) / (2 * a)
x2 <- (-b - sqrt(b**2 - 4*a*c)) / (2 * a)

print(x1)
```

```
[1] 4
```

```r
print(x2)
```

```
[1] 3
```

**Exercise 1.2.** Which are the last 3 values of the intensity column of the dataframe?

Solution

We can do it in many steps:

```
intensity_values <- my_data$intensity

length_intensity <- length(intensity_values)

third_to_last_index <- length_intensity - 2

last_3_intensities <- intensity_values[third_to_last_index:length_intensity]

print(last_3_intensities)
```

```
[1] 22965783 22027811 22338574
```

### 1.3.4 Vector-number operations

What we have seen so far on arithmetical operations between single numbers, can be applied also to numerical vector x single number operations.

In our data, area is expressed in µm^2 and we want to convert it into mm^2. To do so, we should divide each value of the area by a factor of 1,000,000; the cool thing about R is that this is done automatically when we use the statement `numeric_vector <operand> number`:

```
division_factor = 1e6

areas_mm2 <- area_values / division_factor

print(head(areas_mm2))
```

```
[1] 0.003124709 0.003499559 0.003773000 0.003538320 0.003936014 0.003646389
```

```
print(head(area_values))
```

```
[1] 3124.709 3499.559 3773.000 3538.320 3936.014 3646.389
```

This is true for ANY arithmetical operations:

```
print(head(intensity_values))
```

```
[1] 16015465 19449815 21118210 18100075 20718554 19507110
```

```
print(head(intensity_values + 10))
```

```
[1] 16015475 19449825 21118220 18100085 20718564 19507120
```

### 1.3.5 Operations between vectors

It is possible also to perform element-wise operations between vectors.

Let's load a new dataset with some patient data, and calculate the BMI of each patient.

```r
patient_df <- read.csv("data/patient_data.csv")

str(patient_df)
```

```
'data.frame':   100 obs. of  3 variables:
 $ patient_id: int  1 2 3 4 5 6 7 8 9 10 ...
 $ weight    : num  75.8 61.8 91.7 66.4 80.4 ...
 $ height    : num  186 212 188 176 174 ...
```

```r
summary(patient_df)
```

```
   patient_id         weight           height      
 Min.   :  1.00   Min.   : 49.79   Min.   : 91.88  
 1st Qu.: 25.75   1st Qu.: 68.25   1st Qu.:159.20  
 Median : 50.50   Median : 73.91   Median :173.15  
 Mean   : 50.50   Mean   : 75.18   Mean   :174.96  
 3rd Qu.: 75.25   3rd Qu.: 84.61   3rd Qu.:185.90  
 Max.   :100.00   Max.   :112.18   Max.   :253.13  
```

```r
head(patient_df)
```

```
  patient_id   weight   height
1          1 75.82265 185.6147
2          2 61.77137 212.2587
3          3 91.65377 188.3604
4          4 66.38117 176.2986
5          5 80.42689 174.0509
6          6 79.32015 173.7906
```

BMI formula: $\frac{\text{weight in kg}}{(\text{height in m})^2}$

To do so, we should transform the height in m, and then perform the operation. We will save the results as a **new column** of the dataframe; this can be done with the syntax `dataframe_variable$new_column_name <- vector` or `dataframe_variable$["new_column_name"] <- vector` (we will see on day 2 the strings).

```r
# Get height in m
patient_df$height_in_m <- patient_df$height / 100

print(head(patient_df))
```

```
  patient_id   weight   height height_in_m
1          1 75.82265 185.6147    1.856147
2          2 61.77137 212.2587    2.122587
3          3 91.65377 188.3604    1.883604
4          4 66.38117 176.2986    1.762986
5          5 80.42689 174.0509    1.740509
6          6 79.32015 173.7906    1.737906
```

```r
patient_df["BMI"] <- patient_df$weight / (patient_df$height_in_m ** 2)

print(head(patient_df))
```

```
  patient_id   weight   height height_in_m      BMI
1          1 75.82265 185.6147    1.856147 22.00768
2          2 61.77137 212.2587    2.122587 13.71059
3          3 91.65377 188.3604    1.883604 25.83279
4          4 66.38117 176.2986    1.762986 21.35733
5          5 80.42689 174.0509    1.740509 26.54902
6          6 79.32015 173.7906    1.737906 26.26220
```

### 1.3.6 Sumamry statistics of numeric vectors

Usually, when dealing with numeric data we want to have some summary statistics on a specific data (e.g. mean values, median, quartile, standard deviation, …).

In R there are many built-in functions that can help us in doing so:

- `mean` to calculate the mean
- `sd` to calculate the standard deviation
- `variance` to calculate the variance
- `median` to calculate the median
- `quantile` to calculate the nth quantile of a distribution
- `round` to round decimal values to n decimal places
- `sum` to sum all the values of the vector

- many others

They all have in common the syntax: `name_of_the_function(vector)`.

```
intensity_mean <- mean(intensity_values)
print(intensity_mean)
```

```
[1] 19419175
```

```
height_sd <- sd(patient_df$height)
print(height_sd)
```

```
[1] 26.73183
```

```
BMI_median <- median(patient_df$BMI)
print(BMI_median)
```

```
[1] 25.23683
```

```
first_quartile_weight <- quantile(patient_df$weight, probs = 0.25)
print(first_quartile_weight)
```

```
     25%
68.25118
```

```
rounded_BMI <- round(patient_df$BMI, 1)
print(head(rounded_BMI))
```

```
[1] 22.0 13.7 25.8 21.4 26.5 26.3
```

**Exercise 1.3.** Scale BMI values (formula: $scaled_i = \frac{x_i - \overline{x}}{sd(x)}$)

Solution

Here is a step-by-step solution:

```
# calculate BMI mean
mean_BMI <- mean(patient_df$BMI)

# calculate BMI sd
sd_BMI <- sd(patient_df$BMI)

# Calculate scaled BMIs
patient_df$scaled_BMI <- (patient_df$BMI - mean_BMI) / sd_BMI

print(head(patient_df))
```

```
  patient_id   weight   height height_in_m      BMI  scaled_BMI
1          1 75.82265 185.6147    1.856147 22.00768 -0.44542706
2          2 61.77137 212.2587    2.122587 13.71059 -1.29224320
3          3 91.65377 188.3604    1.883604 25.83279 -0.05502959
4          4 66.38117 176.2986    1.762986 21.35733 -0.51180319
5          5 80.42689 174.0509    1.740509 26.54902  0.01807031
6          6 79.32015 173.7906    1.737906 26.26220 -0.01120341
```

That is exactly what the formula `scale` do underneath:

```r
# Calculate scaled BMIs
patient_df$scaled_BMI_function <- scale(patient_df$BMI)

print(head(patient_df))
```

```
  patient_id   weight   height height_in_m      BMI  scaled_BMI
1          1 75.82265 185.6147    1.856147 22.00768 -0.44542706
2          2 61.77137 212.2587    2.122587 13.71059 -1.29224320
3          3 91.65377 188.3604    1.883604 25.83279 -0.05502959
4          4 66.38117 176.2986    1.762986 21.35733 -0.51180319
5          5 80.42689 174.0509    1.740509 26.54902  0.01807031
6          6 79.32015 173.7906    1.737906 26.26220 -0.01120341
  scaled_BMI_function
1         -0.44542706
2         -1.29224320
3         -0.05502959
4         -0.51180319
5          0.01807031
6         -0.01120341
```

## 1.4 Save a dataframe

We are satisfied with this preliminary edit of the patients' data, so we can save the data to a dedicated file.

We use `write.csv` function:

```r
write.csv(patient_df, # what we want to save
          file = "output/patients_data_BMI.csv", # name of the file
          row.names = FALSE) # we'll see row names in day 2
```

## 1.5 Home exercise

For next time, if you want, you can try to do this exercise:

1. Starting from patient data (raw), load the file
2. Inspect it
3. You know that you have to give drug A to each patient so that the final concentration is 10 mg/kg (mg of drug every kg of patient). Calculate how much drug you should give to each patient.
4. Given that a single stock of drug A is 5 g, how many stocks you have to order?

Solution

```r
# 1. Load the file
patient_df <- read.csv("data/patient_data.csv")
```

```r
# 2. Inspect it
head(patient_df)
```

```
  patient_id   weight   height
1          1 75.82265 185.6147
2          2 61.77137 212.2587
3          3 91.65377 188.3604
4          4 66.38117 176.2986
5          5 80.42689 174.0509
6          6 79.32015 173.7906
```

```r
str(patient_df)
```

```
'data.frame':   100 obs. of  3 variables:
 $ patient_id: int  1 2 3 4 5 6 7 8 9 10 ...
 $ weight    : num  75.8 61.8 91.7 66.4 80.4 ...
 $ height    : num  186 212 188 176 174 ...
```

```r
summary(patient_df)
```

```
   patient_id         weight           height      
 Min.   :  1.00   Min.   : 49.79   Min.   : 91.88  
 1st Qu.: 25.75   1st Qu.: 68.25   1st Qu.:159.20  
 Median : 50.50   Median : 73.91   Median :173.15  
 Mean   : 50.50   Mean   : 75.18   Mean   :174.96  
 3rd Qu.: 75.25   3rd Qu.: 84.61   3rd Qu.:185.90  
 Max.   :100.00   Max.   :112.18   Max.   :253.13  
```

```r
# 3. Calculate drug for each patient

patient_df$drugA_quantity_mg <- patient_df$weight * 10
head(patient_df)
```

```
  patient_id   weight   height drugA_quantity_mg
1          1 75.82265 185.6147          758.2265
2          2 61.77137 212.2587          617.7137
3          3 91.65377 188.3604          916.5377
4          4 66.38117 176.2986          663.8117
5          5 80.42689 174.0509          804.2689
6          6 79.32015 173.7906          793.2015
```

```r
# 4. How many stocks you have to order?

total_drug_A_mg <- sum(patient_df$drugA_quantity_mg)

total_drug_A_g <- total_drug_A_mg / 1000

stock_weight <- 5

number_of_stocks <- total_drug_A_g/stock_weight

print(number_of_stocks)
```

```
[1] 15.03539
```

To be precise, we can use `ceiling` function that round to the upper integer:

```r
number_of_stocks <- ceiling(number_of_stocks)

print(number_of_stocks)
```

```
[1] 16
```

# Chapter 2

# Strings, Factors and Dataframe Handling

## 2.1 What we will see today

Today we will look at some patient data, and we will answer these questions:

- Are data consistent?
- Which is the mean and sd values of glucose in each treatment group? How many patients we have per treatment group?
- Which group has the highest mean creatinine? (2 different datasets)

## 2.2 Strings and Factors

Today we will work with a clinical dataset containing information about patients enrolled in a multi-site trial. Let's import it and, as always, **inspect it first**:

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.1     v readr     2.2.0
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.2     v tibble    3.3.1
v lubridate 1.9.5     v tidyr     1.3.2
v purrr     1.2.2
```

```
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to
```

```r
clinical_data <- read.csv("data/clinical_data.csv")
```

```r
str(clinical_data)
```

```
'data.frame':   240 obs. of  15 variables:
 $ patient_id      : chr  "PT-001" "PT-002" "PT-003" "PT-004" ...
 $ site            : chr  "Milan" "Rome" "Rome" "Turin " ...
 $ visit_date      : chr  "2023-07-20" "2020-08-15" "2023-10-15" "2020-03-06" ...
 $ age             : int  40 40 59 46 58 53 51 34 43 60 ...
 $ sex             : chr  "Male" "Male" "Female" "Male" ...
 $ smoking         : chr  "never" "Curent" "former" "Curent" ...
 $ treatment       : chr  "Drug_A" "Drug_B" "Drug_A" "Drug_A" ...
 $ stage           : chr  "II" "I" "I" "I" ...
 $ weight_kg       : num  85.9 100.9 65 95.9 75.7 ...
 $ height_cm       : num  173 185 166 182 168 ...
 $ sbp_mmhg        : int  164 143 152 142 160 150 137 146 151 159 ...
 $ dbp_mmhg        : int  101 102 98 88 96 107 96 106 94 104 ...
 $ glucose_mgdl    : int  113 105 110 120 131 78 108 93 106 110 ...
 $ cholesterol_mgdl: int  172 239 146 232 192 232 163 199 200 185 ...
 $ crp_ngml        : num  5.89 6.13 7.81 3.35 21.72 ...
```

```r
summary(clinical_data)
```

```
  patient_id            site            visit_date             age
 Length:240         Length:240         Length:240         Min.   :26.0
 Class :character   Class :character   Class :character   1st Qu.:49.0
 Mode  :character   Mode  :character   Mode  :character   Median :57.5
                                                          Mean   :57.3
                                                          3rd Qu.:65.0
                                                          Max.   :85.0
     sex              smoking           treatment           stage
 Length:240         Length:240         Length:240         Length:240
 Class :character   Class :character   Class :character   Class :character
 Mode  :character   Mode  :character   Mode  :character   Mode  :character



   weight_kg        height_cm        sbp_mmhg        dbp_mmhg
 Min.   : 39.40   Min.   :148.9   Min.   :115.0   Min.   : 73.00
```

```
 1st Qu.: 65.35   1st Qu.:162.2   1st Qu.:139.0   1st Qu.: 90.75
 Median : 75.65   Median :168.5   Median :146.5   Median : 97.00
 Mean   : 76.87   Mean   :169.2   Mean   :146.7   Mean   : 96.11
 3rd Qu.: 87.83   3rd Qu.:174.8   3rd Qu.:153.2   3rd Qu.:102.00
 Max.   :132.90   Max.   :200.1   Max.   :192.0   Max.   :120.00
  glucose_mgdl   cholesterol_mgdl    crp_ngml
 Min.   : 60.0   Min.   :113.0    Min.   : 1.92
 1st Qu.: 98.0   1st Qu.:189.8    1st Qu.: 5.30
 Median :109.0   Median :207.0    Median : 8.37
 Mean   :108.8   Mean   :207.9    Mean   :10.35
 3rd Qu.:118.0   3rd Qu.:226.0    3rd Qu.:13.02
 Max.   :149.0   Max.   :307.0    Max.   :42.09
```

```r
head(clinical_data)
```

```
  patient_id   site visit_date age    sex smoking treatment stage weight_kg
1     PT-001  Milan 2023-07-20  40   Male   never    Drug_A    II      85.9
2     PT-002   Rome 2020-08-15  40   Male  Curent    Drug_B     I     100.9
3     PT-003   Rome 2023-10-15  59 Female  former    Drug_A     I      65.0
4     PT-004 Turin  2020-03-06  46   Male  Curent    Drug_A     I      95.9
5     PT-005  Turin 2023-03-30  58   Male  FORMER    Drug_B   III      75.7
6     PT-006  Milan 2023-04-15  53 Female  Curent   Control    IV      69.6
  height_cm sbp_mmhg dbp_mmhg glucose_mgdl cholesterol_mgdl crp_ngml
1     172.7      164      101          113              172     5.89
2     185.3      143      102          105              239     6.13
3     165.6      152       98          110              146     7.81
4     181.5      142       88          120              232     3.35
5     168.1      160       96          131              192    21.72
6     174.3      150      107           78              232    32.42
```

You will notice that several columns contain text: `patient_id`, `sex`, `treatment`, `stage`, `smoking`, `site`, `diagnosis_code`. In R, text data can be represented in two fundamentally different ways: as **strings** (also called *characters*) or as **factors**. Let's understand the difference.

---

A **string** (or character) is simply a sequence of text. R treats it as a raw piece of text and does not impose any structure on it.

A **factor** is a *categorical variable*: it has a fixed, predefined set of allowed values called **levels**. Under the hood, R stores factors as integers (one integer per level) and keeps a table that maps integers to labels. This makes factors memory-efficient and very useful for statistical modelling (e.g., in regression,

factors are automatically handled as categorical predictors) and for plotting purposes.

**When to use strings and when factors?**

| Use **string** when… | Use **factor** when… |
|---|---|
| Text is free-form or unpredictable | The variable takes a limited set of known values |
| You need to manipulate the text | Order matters (e.g., stage I < II < III < IV) |
| Unique identifiers (patient IDs) | Input to statistical models |

As some of our columns fall in the conditions for factors, let's convert them with the function `factor`:

```
clinical_data$sex       <- factor(clinical_data$sex)
clinical_data$treatment <- factor(clinical_data$treatment)
clinical_data$site      <- factor(clinical_data$site)
clinical_data$smoking   <- factor(clinical_data$smoking)

# For stage, the categories have a natural ORDER: we declare an ordered factor
clinical_data$stage <- factor(clinical_data$stage,
                              levels = c("I", "II", "III", "IV"),
                              ordered = TRUE)

str(clinical_data)
```

```
'data.frame':   240 obs. of  15 variables:
 $ patient_id      : chr  "PT-001" "PT-002" "PT-003" "PT-004" ...
 $ site            : Factor w/ 4 levels "Milan","Rome",..: 1 2 2 4 3 1 2 2 1 2 ...
 $ visit_date      : chr  "2023-07-20" "2020-08-15" "2023-10-15" "2020-03-06" ...
 $ age             : int  40 40 59 46 58 53 51 34 43 60 ...
 $ sex             : Factor w/ 2 levels "Female","Male": 2 2 1 2 2 1 2 1 2 2 ...
 $ smoking         : Factor w/ 7 levels "Curent","current",..: 5 1 3 1 4 1 3 6 5 4 ...
 $ treatment       : Factor w/ 3 levels "Control","Drug_A",..: 2 3 2 2 3 1 2 1 3 2 ...
 $ stage           : Ord.factor w/ 4 levels "I"<"II"<"III"<..: 2 1 1 1 3 4 1 1 3 2 ...
 $ weight_kg       : num  85.9 100.9 65 95.9 75.7 ...
 $ height_cm       : num  173 185 166 182 168 ...
 $ sbp_mmhg        : int  164 143 152 142 160 150 137 146 151 159 ...
 $ dbp_mmhg        : int  101 102 98 88 96 107 96 106 94 104 ...
 $ glucose_mgdl    : int  113 105 110 120 131 78 108 93 106 110 ...
 $ cholesterol_mgdl: int  172 239 146 232 192 232 163 199 200 185 ...
 $ crp_ngml        : num  5.89 6.13 7.81 3.35 21.72 ...
```

Now `summary` gives us much more useful information on those columns:

```r
summary(clinical_data)
```

```
 patient_id            site     visit_date             age           sex
Length:240         Milan :96   Length:240         Min.   :26.0   Female:118
Class :character   Rome  :77   Class :character   1st Qu.:49.0   Male  :122
Mode  :character   Turin :31   Mode  :character   Median :57.5
                   Turin :36                      Mean   :57.3
                                                  3rd Qu.:65.0
                                                  Max.   :85.0

        smoking     treatment  stage      weight_kg        height_cm
Curent      :35   Control:74   I  :76   Min.   : 39.40   Min.   :148.9
current     :28   Drug_A :88   II :73   1st Qu.: 65.35   1st Qu.:162.2
former      :44   Drug_B :78   III:53   Median : 75.65   Median :168.5
FORMER      :46                IV :38   Mean   : 76.87   Mean   :169.2
never       :26                         3rd Qu.: 87.83   3rd Qu.:174.8
Never       :28                         Max.   :132.90   Max.   :200.1
never smoker:33
   sbp_mmhg        dbp_mmhg       glucose_mgdl   cholesterol_mgdl
Min.   :115.0   Min.   : 73.00   Min.   : 60.0   Min.   :113.0
1st Qu.:139.0   1st Qu.: 90.75   1st Qu.: 98.0   1st Qu.:189.8
Median :146.5   Median : 97.00   Median :109.0   Median :207.0
Mean   :146.7   Mean   : 96.11   Mean   :108.8   Mean   :207.9
3rd Qu.:153.2   3rd Qu.:102.00   3rd Qu.:118.0   3rd Qu.:226.0
Max.   :192.0   Max.   :120.00   Max.   :149.0   Max.   :307.0

   crp_ngml
Min.   : 1.92
1st Qu.: 5.30
Median : 8.37
Mean   :10.35
3rd Qu.:13.02
Max.   :42.09
```

Notice how `summary` now shows the count per level for factors (e.g. how many Male vs Female), while before it just said `character`.

Moreover, we can see that in smoking column, we have different labels with the same meaning (never, Never, never smoker for never). R (as well as any programming language) is case sensitive, so it treats those 3 as different labels, even if for us they mean the same thing.

**IMPORTANT**: labels should be consistent across datasets when you fill them.

As we want to edit those to be consistent, we need to change the column back to be a character, with `as.character` function:

```
clinical_data$smoking   <- as.character(clinical_data$smoking)
str(clinical_data)
```

```
'data.frame':   240 obs. of  15 variables:
 $ patient_id      : chr  "PT-001" "PT-002" "PT-003" "PT-004" ...
 $ site            : Factor w/ 4 levels "Milan","Rome",..: 1 2 2 4 3 1 2 2 1 2 ...
 $ visit_date      : chr  "2023-07-20" "2020-08-15" "2023-10-15" "2020-03-06" ...
 $ age             : int  40 40 59 46 58 53 51 34 43 60 ...
 $ sex             : Factor w/ 2 levels "Female","Male": 2 2 1 2 2 1 2 1 2 2 ...
 $ smoking         : chr  "never" "Curent" "former" "Curent" ...
 $ treatment       : Factor w/ 3 levels "Control","Drug_A",..: 2 3 2 2 3 1 2 1 3 2 ...
 $ stage           : Ord.factor w/ 4 levels "I"<"II"<"III"<..: 2 1 1 1 3 4 1 1 3 2 ...
 $ weight_kg       : num  85.9 100.9 65 95.9 75.7 ...
 $ height_cm       : num  173 185 166 182 168 ...
 $ sbp_mmhg        : int  164 143 152 142 160 150 137 146 151 159 ...
 $ dbp_mmhg        : int  101 102 98 88 96 107 96 106 94 104 ...
 $ glucose_mgdl    : int  113 105 110 120 131 78 108 93 106 110 ...
 $ cholesterol_mgdl: int  172 239 146 232 192 232 163 199 200 185 ...
 $ crp_ngml        : num  5.89 6.13 7.81 3.35 21.72 ...
```

We can now proceed.

### 2.2.1 Operations on strings

The main package for string operations in R (and the one we will use throughout the course) is `stringr`, which is part of the `tidyverse`. All its functions start with `str_` to make them easy to remember.

#### 2.2.1.1 Character substitution

Let's start by looking at how to change characters in a string: from "Curent" to "current" using `str_replace(vector, old_pattern, new_pattern)`.

```
print(unique(clinical_data$smoking))
```

```
[1] "never"        "Curent"       "former"       "FORMER"       "Never"
[6] "never smoker" "current"
```

```
clinical_data$smoking <- str_replace(clinical_data$smoking, "Curent", "current")
print(unique(clinical_data$smoking))
```

```
[1] "never"        "current"      "former"       "FORMER"       "Never"
[6] "never smoker"
```

Then we can unify "FORMER" and "former" by transforming the former to lowercase with `str_to_lower`:

```
print(unique(clinical_data$smoking))
```

```
[1] "never"        "current"      "former"       "FORMER"       "Never"
[6] "never smoker"
```

```
clinical_data$smoking <- str_to_lower(clinical_data$smoking)
print(unique(clinical_data$smoking))
```

```
[1] "never"        "current"      "former"       "never smoker"
```

We have converted both FORMER to former and Never to never. We lastly have to change "never smoker" to "never".

**Exercise 2.1.** Change "never smoker" to "smoker" and make smoker as factor.

Solution

```
clinical_data$smoking <- str_replace(clinical_data$smoking, "never smoker", "never")
clinical_data$smoking <- factor(clinical_data$smoking, levels = c("never", "former", "current"))

summary(clinical_data)
```

```
  patient_id            site     visit_date             age           sex
 Length:240         Milan :96   Length:240         Min.   :26.0   Female:118
 Class :character   Rome  :77   Class :character   1st Qu.:49.0   Male  :122
 Mode  :character   Turin :31   Mode  :character   Median :57.5
                    Turin :36                      Mean   :57.3
                                                   3rd Qu.:65.0
                                                   Max.   :85.0
    smoking     treatment  stage      weight_kg        height_cm
 never  :87   Control:74   I  :76   Min.   : 39.40   Min.   :148.9
 former :90   Drug_A :88   II :73   1st Qu.: 65.35   1st Qu.:162.2
 current:63   Drug_B :78   III:53   Median : 75.65   Median :168.5
                           IV :38   Mean   : 76.87   Mean   :169.2
                                    3rd Qu.: 87.83   3rd Qu.:174.8
                                    Max.   :132.90   Max.   :200.1
    sbp_mmhg        dbp_mmhg       glucose_mgdl   cholesterol_mgdl
```

```
 Min.   :115.0   Min.   : 73.00   Min.   : 60.0   Min.   :113.0
 1st Qu.:139.0   1st Qu.: 90.75   1st Qu.: 98.0   1st Qu.:189.8
 Median :146.5   Median : 97.00   Median :109.0   Median :207.0
 Mean   :146.7   Mean   : 96.11   Mean   :108.8   Mean   :207.9
 3rd Qu.:153.2   3rd Qu.:102.00   3rd Qu.:118.0   3rd Qu.:226.0
 Max.   :192.0   Max.   :120.00   Max.   :149.0   Max.   :307.0
    crp_ngml
 Min.   : 1.92
 1st Qu.: 5.30
 Median : 8.37
 Mean   :10.35
 3rd Qu.:13.02
 Max.   :42.09
```

For sake of completeness, we could have done all of this with a single function `replace_values` (for new versions of R):

```
clinical_data$smoking <- replace_values(as.character(clinical_data$smoking), c("never s
                                        c("Curent") ~ "current",
                                        c("FORMER") ~ "former")

clinical_data$smoking = factor(clinical_data$smoking)
```

If it doesn't work, you could use:

```
clinical_data = mutate(smoking = case_when(
    smoking %in% c("never smoker", "Never") ~ "never",
    smoking %in% c("FORMER", "Former", "Fomer") ~ "former",
    smoking == "Curent" ~ "current"),
    .data = clinical_data
    )

clinical_data$smoking = factor(clinical_data$smoking)
```

Q: *Why there are 2 Turin?* A: Let's check in details the possible labels of that column with function `levels`:

```
print(levels(clinical_data$site))
```

```
[1] "Milan"  "Rome"   "Turin"  "Turin "
```

Here's the hitch, some have a trailing space after Turin. We can get rid of them using `str_trim` function:

```r
print(levels(clinical_data$site))
```

```
[1] "Milan"  "Rome"   "Turin"  "Turin "
```

```r
clinical_data$site <- factor(str_trim(clinical_data$site))

print(levels(clinical_data$site))
```

```
[1] "Milan" "Rome"  "Turin"
```

```r
print(summary(clinical_data))
```

```
  patient_id           site     visit_date             age           sex
 Length:240         Milan:96   Length:240         Min.   :26.0   Female:118
 Class :character   Rome :77   Class :character   1st Qu.:49.0   Male  :122
 Mode  :character   Turin:67   Mode  :character   Median :57.5
                                                   Mean   :57.3
                                                   3rd Qu.:65.0
                                                   Max.   :85.0
 smoking      treatment  stage      weight_kg        height_cm
 NA's:240   Control:74   I  :76   Min.   : 39.40   Min.   :148.9
            Drug_A :88   II :73   1st Qu.: 65.35   1st Qu.:162.2
            Drug_B :78   III:53   Median : 75.65   Median :168.5
                         IV :38   Mean   : 76.87   Mean   :169.2
                                  3rd Qu.: 87.83   3rd Qu.:174.8
                                  Max.   :132.90   Max.   :200.1
    sbp_mmhg        dbp_mmhg       glucose_mgdl   cholesterol_mgdl
 Min.   :115.0   Min.   : 73.00   Min.   : 60.0   Min.   :113.0
 1st Qu.:139.0   1st Qu.: 90.75   1st Qu.: 98.0   1st Qu.:189.8
 Median :146.5   Median : 97.00   Median :109.0   Median :207.0
 Mean   :146.7   Mean   : 96.11   Mean   :108.8   Mean   :207.9
 3rd Qu.:153.2   3rd Qu.:102.00   3rd Qu.:118.0   3rd Qu.:226.0
 Max.   :192.0   Max.   :120.00   Max.   :149.0   Max.   :307.0
    crp_ngml
 Min.   : 1.92
 1st Qu.: 5.30
 Median : 8.37
 Mean   :10.35
 3rd Qu.:13.02
 Max.   :42.09
```

#### 2.2.1.2 Other useful functions to handle strings

A few more `stringr` functions you will find very useful:

| Function | What it does |
|---|---|
| `str_detect(x, pattern)` | Returns TRUE/FALSE: does the string contain the pattern? |
| `str_starts(x, pattern)` | Does the string *start* with the pattern? |
| `str_ends(x, pattern)` | Does the string *end* with the pattern? |
| `str_pad(x, width, side, pad)` | Pads a string to a given width |
| `str_sub(x, start, end)` | Extracts a substring by position |

---

### 2.2.2 Operations on multiple columns

#### 2.2.2.1 Split a column and merge multiple columns

Sometimes a single column contains multiple pieces of information that could be split into separate columns. Look at `visit_date`:

```
head(clinical_data$visit_date)
```

```
[1] "2023-07-20" "2020-08-15" "2023-10-15" "2020-03-06" "2023-03-30"
[6] "2023-04-15"
```

We can see that the year of visit varies a lot, maybe we should take it into account. We can split this into 3 separate columns using `separate` function:

```
clinical_data <- separate(clinical_data,
                          col = visit_date,         # column to split
                          into = c("visit_year", "visit_month", "visit_day"), # new c
                          sep = "-",                # separator
                          remove = FALSE)           # keep the original column

head(clinical_data)
```

```
  patient_id  site visit_date visit_year visit_month visit_day age    sex
1     PT-001 Milan 2023-07-20       2023          07        20  40   Male
2     PT-002  Rome 2020-08-15       2020          08        15  40   Male
3     PT-003  Rome 2023-10-15       2023          10        15  59 Female
4     PT-004 Turin 2020-03-06       2020          03        06  46   Male
5     PT-005 Turin 2023-03-30       2023          03        30  58   Male
6     PT-006 Milan 2023-04-15       2023          04        15  53 Female
  smoking treatment stage weight_kg height_cm sbp_mmhg dbp_mmhg glucose_mgdl
```

```
1    <NA>    Drug_A    II      85.9     172.7      164      101          113
2    <NA>    Drug_B     I     100.9     185.3      143      102          105
3    <NA>    Drug_A     I      65.0     165.6      152       98          110
4    <NA>    Drug_A     I      95.9     181.5      142       88          120
5    <NA>    Drug_B   III      75.7     168.1      160       96          131
6    <NA>   Control    IV      69.6     174.3      150      107           78
  cholesterol_mgdl crp_ngml
1              172     5.89
2              239     6.13
3              146     7.81
4              232     3.35
5              192    21.72
6              232    32.42
```

The opposite operation — merging multiple columns into one — is done with `unite`:

```
# example: unite back together
clinical_data <- unite(clinical_data,
                       col = "visit_year_month",         # name of the new column
                       visit_year, visit_month,           # columns to merge
                       sep = "-",                       # separator
                       remove = FALSE)                  # keep original columns

head(clinical_data)
```

```
  patient_id  site visit_date visit_year_month visit_year visit_month visit_day
1     PT-001 Milan 2023-07-20          2023-07       2023          07        20
2     PT-002  Rome 2020-08-15          2020-08       2020          08        15
3     PT-003  Rome 2023-10-15          2023-10       2023          10        15
4     PT-004 Turin 2020-03-06          2020-03       2020          03        06
5     PT-005 Turin 2023-03-30          2023-03       2023          03        30
6     PT-006 Milan 2023-04-15          2023-04       2023          04        15
  age    sex smoking treatment stage weight_kg height_cm sbp_mmhg dbp_mmhg
1  40   Male    <NA>    Drug_A    II      85.9     172.7      164      101
2  40   Male    <NA>    Drug_B     I     100.9     185.3      143      102
3  59 Female    <NA>    Drug_A     I      65.0     165.6      152       98
4  46   Male    <NA>    Drug_A     I      95.9     181.5      142       88
5  58   Male    <NA>    Drug_B   III      75.7     168.1      160       96
6  53 Female    <NA>   Control    IV      69.6     174.3      150      107
  glucose_mgdl cholesterol_mgdl crp_ngml
1          113              172     5.89
2          105              239     6.13
3          110              146     7.81
4          120              232     3.35
```

```
5          131              192    21.72
6           78              232    32.42
```

---

### 2.2.3 Column and row names

Every dataframe has **column names** (the variable names) and **row names** (by default a number, but we can change it).

To retrieve column names:

```r
colnames(clinical_data)
```

```
 [1] "patient_id"       "site"             "visit_date"       "visit_year_month"
 [5] "visit_year"       "visit_month"      "visit_day"        "age"
 [9] "sex"              "smoking"          "treatment"        "stage"
[13] "weight_kg"        "height_cm"        "sbp_mmhg"         "dbp_mmhg"
[17] "glucose_mgdl"     "cholesterol_mgdl" "crp_ngml"
```

To rename specific columns, we can assign a new vector of names to `colnames`, or use the tidyverse function `rename` (we will see this with `dplyr` below):

```r
clinical_data = rename(clinical_data, "diastolic_bp_mmhg" = "dbp_mmhg", "systolic_bp_mm

colnames(clinical_data)
```

```
 [1] "patient_id"        "site"              "visit_date"
 [4] "visit_year_month"  "visit_year"        "visit_month"
 [7] "visit_day"         "age"               "sex"
[10] "smoking"           "treatment"         "stage"
[13] "weight_kg"         "height_cm"         "systolic_bp_mmhg"
[16] "diastolic_bp_mmhg" "glucose_mgdl"      "cholesterol_mgdl"
[19] "crp_ngml"
```

For row names:

```r
# by default, row names are row numbers
head(rownames(clinical_data))
```

```
[1] "1" "2" "3" "4" "5" "6"
```

```
# we could set patient_id as row names
rownames(clinical_data) <- clinical_data$patient_id

head(rownames(clinical_data))
```

```
[1] "PT-001" "PT-002" "PT-003" "PT-004" "PT-005" "PT-006"
```

However, **be careful** with row names: they are not a proper column of the dataframe, so they don't follow along easily in many tidyverse operations. It is generally better to keep your identifiers as a regular column.

---

## 2.3 Dataframe Handling

The main verbs we will learn today are:

- `mutate` — add or modify columns
- `select` — choose which columns to keep
- `filter`/`filter_out` — keep/remove rows that satisfy a condition
- `group_by` + `summarize` / `reframe` — aggregate data by groups
- `left_join` / `inner_join` — merge two dataframes

### 2.3.1 Create new columns with mutate

`mutate` creates new columns or modifies existing ones.

A key feature of `dplyr` is the **pipe operator** `|>` (or `%>%` in older tidyverse style). It passes the result of one operation directly to the next, making code much more readable:

You can read `|>` as *"and then"*.

```
# Add a BMI column (we already know this formula!)
clinical_data <- clinical_data |> # take clinical_data and then
  mutate(height_m = height_cm / 100,
         BMI = weight_kg / height_m^2,
         BMI = round(BMI, 1))

head(clinical_data)
```

```
       patient_id  site visit_date visit_year_month visit_year visit_month
PT-001     PT-001 Milan 2023-07-20          2023-07       2023          07
PT-002     PT-002  Rome 2020-08-15          2020-08       2020          08
PT-003     PT-003  Rome 2023-10-15          2023-10       2023          10
PT-004     PT-004 Turin 2020-03-06          2020-03       2020          03
PT-005     PT-005 Turin 2023-03-30          2023-03       2023          03
PT-006     PT-006 Milan 2023-04-15          2023-04       2023          04
       visit_day age    sex smoking treatment stage weight_kg height_cm
PT-001        20  40   Male    <NA>    Drug_A    II      85.9     172.7
PT-002        15  40   Male    <NA>    Drug_B     I     100.9     185.3
PT-003        15  59 Female    <NA>    Drug_A     I      65.0     165.6
PT-004        06  46   Male    <NA>    Drug_A     I      95.9     181.5
PT-005        30  58   Male    <NA>    Drug_B   III      75.7     168.1
PT-006        15  53 Female    <NA>   Control    IV      69.6     174.3
       systolic_bp_mmhg diastolic_bp_mmhg glucose_mgdl cholesterol_mgdl
PT-001              164               101          113              172
PT-002              143               102          105              239
PT-003              152                98          110              146
PT-004              142                88          120              232
PT-005              160                96          131              192
PT-006              150               107           78              232
       crp_ngml height_m  BMI
PT-001     5.89    1.727 28.8
PT-002     6.13    1.853 29.4
PT-003     7.81    1.656 23.7
PT-004     3.35    1.815 29.1
PT-005    21.72    1.681 26.8
PT-006    32.42    1.743 22.9
```

`mutate` can also create columns based on conditions, using `if_else` or `case_when`:

```r
# Classify BMI into categories
clinical_data <- clinical_data |>
  mutate(BMI_class = case_when(
    BMI < 18.5 ~ "Underweight",
    BMI < 25   ~ "Normal",
    BMI < 30   ~ "Overweight",
    BMI >= 30  ~ "Obese"
  ),
  BMI_class = factor(BMI_class, labels = c("Underweight", "Normal", "Overweight", "Obes
  )

summary(clinical_data$BMI_class)
```

```
Underweight      Normal  Overweight       Obese
         67          58         102          13
```

---

### 2.3.2 select

`select` chooses which **columns** to keep (or drop). This is useful when working with wide dataframes where you only need a subset of variables.

```r
# keep only demographic and clinical variables
demo_cols <- clinical_data |>
  select(patient_id, site, age, sex, treatment, stage, BMI, BMI_class)

head(demo_cols)
```

```
       patient_id  site age    sex treatment stage  BMI   BMI_class
PT-001     PT-001 Milan  40   Male    Drug_A    II 28.8  Overweight
PT-002     PT-002  Rome  40   Male    Drug_B     I 29.4  Overweight
PT-003     PT-003  Rome  59 Female    Drug_A     I 23.7 Underweight
PT-004     PT-004 Turin  46   Male    Drug_A     I 29.1  Overweight
PT-005     PT-005 Turin  58   Male    Drug_B   III 26.8  Overweight
PT-006     PT-006 Milan  53 Female   Control    IV 22.9 Underweight
```

To **drop** specific columns, use `-` before the column name:

```r
colnames(clinical_data)
```

```
 [1] "patient_id"        "site"              "visit_date"
 [4] "visit_year_month"  "visit_year"        "visit_month"
 [7] "visit_day"         "age"               "sex"
[10] "smoking"           "treatment"         "stage"
[13] "weight_kg"         "height_cm"         "systolic_bp_mmhg"
[16] "diastolic_bp_mmhg" "glucose_mgdl"      "cholesterol_mgdl"
[19] "crp_ngml"          "height_m"          "BMI"
[22] "BMI_class"
```

```r
# drop the helper/intermediate columns we created
clinical_data <- clinical_data |>
  select(-visit_year_month)

colnames(clinical_data)
```

```
 [1] "patient_id"        "site"              "visit_date"
 [4] "visit_year"        "visit_month"       "visit_day"
 [7] "age"               "sex"               "smoking"
[10] "treatment"         "stage"             "weight_kg"
[13] "height_cm"         "systolic_bp_mmhg"  "diastolic_bp_mmhg"
[16] "glucose_mgdl"      "cholesterol_mgdl"  "crp_ngml"
[19] "height_m"          "BMI"               "BMI_class"
```

`select` also supports handy helper functions:

```r
# select all columns that start with "diag"
clinical_data |> select(starts_with("visit")) |> head()
```

```
       visit_date visit_year visit_month visit_day
PT-001 2023-07-20       2023          07        20
PT-002 2020-08-15       2020          08        15
PT-003 2023-10-15       2023          10        15
PT-004 2020-03-06       2020          03        06
PT-005 2023-03-30       2023          03        30
PT-006 2023-04-15       2023          04        15
```

```r
# select all columns that contain "bp"
clinical_data |> select(contains("bp")) |> head()
```

```
       systolic_bp_mmhg diastolic_bp_mmhg
PT-001              164               101
PT-002              143               102
PT-003              152                98
PT-004              142                88
PT-005              160                96
PT-006              150               107
```

---

### 2.3.3 Boolean Operators

A **boolean** (or logical) is a data type that can only take two values: `TRUE` or `FALSE`. It is the backbone of filtering and conditional logic in R.

**A boolean is the answer to a yes/no question.**

For example: *"Is this patient in stage III or IV?"* → `TRUE` or `FALSE`.

#### 2.3.3.1 Comparison operators

These operators compare two values and return a boolean:

| Operator | Meaning | Example | Result |
|---|---|---|---|
| `==` | equal to | `3 == 3` | `TRUE` |
| `!=` | not equal to | `3 != 4` | `TRUE` |
| `>` | greater than | `5 > 3` | `TRUE` |
| `<` | less than | `2 < 1` | `FALSE` |
| `>=` | greater or equal | `3 >= 3` | `TRUE` |
| `<=` | less or equal | `2 <= 1` | `FALSE` |

```r
# simple examples
print(10 > 5)
```

```
[1] TRUE
```

```r
print(10 == 10)
```

```
[1] TRUE
```

```r
print(10 != 10)
```

```
[1] FALSE
```

```r
print(3.14 >= 3.14)
```

```
[1] TRUE
```

When applied to a **vector**, the comparison is done **element-wise** and returns a boolean vector:

```r
ages <- clinical_data$age
head(ages)
```

```
[1] 40 40 59 46 58 53
```

```r
is_over_60 <- ages > 60
head(is_over_60)
```

```
[1] FALSE FALSE FALSE FALSE FALSE FALSE
```

Under the hood, TRUE is stored as a 1, while FALSE as 0. This is useful when we want to get the number of TRUE values of a vector. How? By summing up all the values:

```r
sum(is_over_60)
```

```
[1] 84
```

#### 2.3.3.2 Logical operators

We often need to combine multiple conditions. This is done with **logical operators**:

| Operator | Meaning | Example |
|---|---|---|
| `&` | AND — both conditions must be true | `age > 50 & sex == "Female"` |
| `\|` | OR — at least one condition must be true | `stage == "III" \| stage == "IV"` |
| `!` | NOT — negates a condition | `!is.na(crp_ngml)` |

```r
# patients who are Female AND over 60
female_over60 <- (clinical_data$sex == "Female") & (clinical_data$age > 60)
sum(female_over60)
```

```
[1] 49
```

```r
# patients in advanced stage (III or IV)
advanced_stage <- (clinical_data$stage == "III") | (clinical_data$stage == "IV")
sum(advanced_stage)
```

```
[1] 91
```

```r
# NOT: patients who are NOT in the Control group
not_control <- clinical_data$treatment != "Control"
sum(not_control)
```

```
[1] 166
```

```r
# equivalent using !
not_control_2 <- !(clinical_data$treatment == "Control")
sum(not_control_2)
```

```
[1] 166
```

A very useful operator when checking membership in a list of values is `%in%`:

```r
# patients from Milan or Rome (equivalent to == "Milan" | == "Rome")
big_sites <- clinical_data$site %in% c("Milan", "Rome")
sum(big_sites)
```

```
[1] 173
```

::: .{exercise #bool-sum}

How many patients have been diagnosed as stage I, in Turin in 2018?

:::

Solution

```r
stageI_turin_2018 <- (clinical_data$site == "Turin") & (clinical_data$stage == "I") & (clinical_d
sum(stageI_turin_2018)
```

```
[1] 4
```

### 2.3.4 filter and filter out

`filter` keeps only the **rows** that satisfy one or more conditions:

```r
# Keep only patients treated with Drug_A
drug_a_patients <- clinical_data |>
  filter(treatment == "Drug_A")

nrow(drug_a_patients)
```

```
[1] 88
```

```r
head(drug_a_patients)
```

```
       patient_id  site visit_date visit_year visit_month visit_day age    sex
PT-001     PT-001 Milan 2023-07-20       2023          07        20  40   Male
PT-003     PT-003  Rome 2023-10-15       2023          10        15  59 Female
PT-004     PT-004 Turin 2020-03-06       2020          03        06  46   Male
PT-007     PT-007  Rome 2020-09-09       2020          09        09  51   Male
PT-010     PT-010  Rome 2023-09-27       2023          09        27  60   Male
PT-012     PT-012  Rome 2025-03-20       2025          03        20  52   Male
       smoking treatment stage weight_kg height_cm systolic_bp_mmhg
PT-001    <NA>    Drug_A    II      85.9     172.7              164
PT-003    <NA>    Drug_A     I      65.0     165.6              152
PT-004    <NA>    Drug_A     I      95.9     181.5              142
PT-007    <NA>    Drug_A     I      69.7     169.7              137
PT-010    <NA>    Drug_A    II      93.4     186.3              159
PT-012    <NA>    Drug_A   III      84.3     168.8              147
       diastolic_bp_mmhg glucose_mgdl cholesterol_mgdl crp_ngml height_m  BMI
PT-001               101          113              172     5.89    1.727 28.8
PT-003                98          110              146     7.81    1.656 23.7
PT-004                88          120              232     3.35    1.815 29.1
PT-007                96          108              163     5.41    1.697 24.2
PT-010               104          110              185     5.30    1.863 26.9
PT-012                96           86              222    10.45    1.688 29.6
         BMI_class
PT-001  Overweight
PT-003 Underweight
PT-004  Overweight
PT-007 Underweight
PT-010  Overweight
PT-012  Overweight
```

```r
# Patients in advanced stage AND not in control group
advanced_treated <- clinical_data |>
  filter((stage == "III" | stage == "IV") & treatment != "Control")

nrow(advanced_treated)
```

```
[1] 76
```

```r
# Using %in% for multiple values
milan_or_rome <- clinical_data |>
  filter(site %in% c("Milan", "Rome"))

nrow(milan_or_rome)
```

```
[1] 173
```

To **filter out** (i.e., *exclude* rows matching a condition), use `filter_out` (for new versions of R and tidyverse):

```r
# Exclude patients from Naples
no_milan <- clinical_data |>
  filter_out(site %in% c("Milan"))

nrow(no_milan)
```

```
[1] 144
```

If it doesn't work, you could use:

```r
no_milan = clinical_data %>%
    filter(!(site == "Milan"))

nrow(no_milan)
```

```
[1] 144
```

Pipes shine when you chain multiple operations:

```r
# Female patients, over 55, from Milan, keep only key columns
subset_females <- clinical_data |>
  filter(sex == "Female", age > 55, site == "Milan") |>
  select(patient_id, age, stage, treatment, BMI, glucose_mgdl)

head(subset_females)
```

```
       patient_id age stage treatment  BMI glucose_mgdl
PT-011     PT-011  65    II   Control 30.5           94
PT-014     PT-014  71     I   Control 21.5           86
PT-018     PT-018  72   III    Drug_A 25.2           83
PT-037     PT-037  58    II    Drug_B 26.8          108
PT-043     PT-043  60   III    Drug_B 25.9          109
PT-045     PT-045  65    II   Control 27.8           97
```

```r
nrow(subset_females)
```

```
[1] 27
```

Note: inside `filter`, multiple conditions separated by commas are treated as AND.

**Exercise 2.2.** From the full dataset, extract patients who:

- are in stage II or III,
- have a glucose level between 90 and 130 mg/dL (inclusive),
- are assigned to Drug_A or Drug_B.

Keep only `patient_id`, `site`, `stage`, `treatment`, and `glucose_mgdl`.

Solution

```r
exercise_result <- clinical_data |>
  filter(stage %in% c("II", "III"),
         glucose_mgdl >= 90,
         glucose_mgdl <= 130,
         treatment %in% c("Drug_A", "Drug_B")) |>
  select(patient_id, site, stage, treatment, glucose_mgdl)

head(exercise_result)
```

```
       patient_id  site stage treatment glucose_mgdl
PT-001     PT-001 Milan    II    Drug_A          113
PT-009     PT-009 Milan   III    Drug_B          106
PT-010     PT-010  Rome    II    Drug_A          110
PT-016     PT-016 Turin    II    Drug_B           96
PT-020     PT-020 Turin   III    Drug_B           98
PT-026     PT-026 Turin   III    Drug_A          122
```

```r
nrow(exercise_result)
```

```
[1] 68
```

---

### 2.3.5 group_by, summarize and reframe

One of the most powerful operations in data analysis is **grouping** data and computing summaries per group. In R, this is done with `group_by` + `summarize`.

**How it works:**

1. `group_by(column)` tells R: *"split the dataframe into groups based on the values of this column"*
2. `summarize(...)` computes one summary value per group

```
Original data:                 After group_by(treatment):

patient  treatment  glucose       GROUP: Control
PT-001   Control    105           PT-001  Control  105
PT-002   Drug_A     118    --->   PT-007  Control  98
PT-003   Drug_B     97            ...
PT-007   Control    98     --->   GROUP: Drug_A
...                               PT-002  Drug_A  118
                                  ...
```

We can perform several operation with this combo, and answer many questions. For example "*Which is the mean and sd values of glucose in each treatment group? How many patients we have per treatment group?*".

```r
# Mean glucose per treatment group
glucose_by_treatment <- clinical_data |>
  group_by(treatment) |>
  summarize(mean_glucose = mean(glucose_mgdl),
            sd_glucose   = sd(glucose_mgdl),
            n_patients   = n())

glucose_by_treatment
```

```
# A tibble: 3 x 4
  treatment mean_glucose sd_glucose n_patients
  <fct>            <dbl>      <dbl>      <int>
1 Control           109.       15.9         74
2 Drug_A            110.       14.9         88
3 Drug_B            108.       16.0         78
```

You can also group by multiple columns at once.

"*For each treatment/sex combination, which is the mean BMI and the median of the age?*"

```r
# Summary by treatment AND sex
summary_by_group <- clinical_data |>
  group_by(treatment, sex) |>
  summarize(mean_BMI    = round(mean(BMI), 2),
            median_age  = median(age),
            n           = n(),
```

```
            .groups = "drop")   # good practice: drop grouping after summarize

summary_by_group
```

```
# A tibble: 6 x 5
  treatment sex    mean_BMI median_age     n
  <fct>     <fct>     <dbl>      <dbl> <int>
1 Control   Female     27.1       59      39
2 Control   Male       28.5       56      35
3 Drug_A    Female     25.6       58      43
4 Drug_A    Male       27.2       55      45
5 Drug_B    Female     25.2       59.5    36
6 Drug_B    Male       26.8       56      42
```

`summarize` returns **one row per group**. If you need to return **multiple rows per group** (e.g., all quantiles), use `reframe` instead:

```
# Get quartiles of CRP per treatment group
crp_quantiles <- clinical_data |>
  group_by(treatment) |>
  reframe(quantile_value = quantile(crp_ngml, probs = c(0.25, 0.5, 0.75)),
          quantile_name  = c("Q1", "median", "Q3"))

crp_quantiles
```

```
# A tibble: 9 x 3
  treatment quantile_value quantile_name
  <fct>              <dbl> <chr>
1 Control             5.18 Q1
2 Control             8.39 median
3 Control            13.8  Q3
4 Drug_A              5.25 Q1
5 Drug_A              7.3  median
6 Drug_A             11.4  Q3
7 Drug_B              5.36 Q1
8 Drug_B              8.52 median
9 Drug_B             12.7  Q3
```

Another useful function: `count`, which is a shortcut for `group_by` + `summarize(n = n())`:

```
# Count patients per site and stage
clinical_data |>
  count(site, stage, sort = TRUE)
```

```
    site stage  n
1  Milan    II 35
2   Rome     I 33
3  Milan     I 23
4  Turin    II 22
5   Rome   III 21
6  Turin     I 20
7  Milan   III 19
8  Milan    IV 19
9   Rome    II 16
10 Turin   III 13
11 Turin    IV 12
12  Rome    IV  7
```

**Exercise 2.3.** For each combination of `site` and `sex`, compute: - the number of patients, - the mean systolic blood pressure, - the proportion of hypertensive patients (hint: hypertension is when systolic blood pression is > 140).

Solution

```r
bp_summary <- clinical_data |>
    mutate(hypertension = systolic_bp_mmhg > 140) |>
    group_by(site, sex) |>
    summarize(
        n_patients          = n(),
        mean_systolic_bp_mmhg    = round(mean(systolic_bp_mmhg), 1),
        prop_hypertensive   = round(mean(hypertension), 3),
        .groups = "drop"
    )

bp_summary
```

```
# A tibble: 6 x 5
  site  sex    n_patients mean_systolic_bp_mmhg prop_hypertensive
  <fct> <fct>       <int>                 <dbl>             <dbl>
1 Milan Female         48                  146.             0.708
2 Milan Male           48                  149.             0.75
3 Rome  Female         39                  144.             0.667
4 Rome  Male           38                  148.             0.763
5 Turin Female         31                  143.             0.581
6 Turin Male           36                  150.             0.889
```

### 2.3.6 Join / Merge

In real-world analyses, data about the same subjects often lives in **separate files**: demographic data here, lab results there, outcomes somewhere else. To combine them, we **join** them on a shared key column (usually a patient ID).

Let's load the separate lab results file:

```
lab_data <- read.csv("data/lab_results.csv")
str(lab_data)
```

```
'data.frame':   160 obs. of  6 variables:
 $ patient_id     : chr  "PT-002" "PT-003" "PT-004" "PT-005" ...
 $ hemoglobin_gdl : num  15 11.8 16 15.4 14 16.2 10.5 15.6 12.4 14.5 ...
 $ wbc_10e9l      : num  7.49 7.46 7.09 6.63 6.99 9.26 4.79 5.65 8.39 7.87 ...
 $ platelets_10e9l: int  270 252 233 192 195 188 163 269 165 263 ...
 $ alt_ul         : int  56 27 20 16 61 51 50 18 47 51 ...
 $ creatinine_mgdl: num  1.58 1.09 1.17 1.06 1.15 1.06 1.21 1.08 1.12 1.03 ...
```

```
head(lab_data)
```

```
  patient_id hemoglobin_gdl wbc_10e9l platelets_10e9l alt_ul creatinine_mgdl
1     PT-002           15.0      7.49             270     56            1.58
2     PT-003           11.8      7.46             252     27            1.09
3     PT-004           16.0      7.09             233     20            1.17
4     PT-005           15.4      6.63             192     16            1.06
5     PT-007           14.0      6.99             195     61            1.15
6     PT-008           16.2      9.26             188     51            1.06
```

The key column is `patient_id`. Not all patients have lab results. Depending on how we want to handle this, we choose different types of join:

| Join type | What it does |
|---|---|
| `left_join(x, y)` | Keep **all rows from x**, add y columns where available (NA if no match) |
| `right_join(x, y)` | Keep all rows from y |
| `inner_join(x, y)` | Keep only rows that exist in **both** x and y |
| `full_join(x, y)` | Keep all rows from both, NA where no match |

Let's see all approaches:

```r
# left_join: keep all patients, add lab data where available
left_df <- clinical_data |>
  left_join(lab_data, by = "patient_id")

nrow(left_df)
```

```
[1] 240
```

```r
# patients without lab data will have NA in lab columns
sum(is.na(left_df$hemoglobin_gdl))
```

```
[1] 80
```

Function `is.na()` returns TRUE when a value is NA, and FALSE if a value is an acutal value. Here an example:

```r
v1 = c(1, NA, 4) # create a vector with one NA

is.na(v1) # returns a vector with TRUE in position 2, and FALSE elsewhere
```

```
[1] FALSE  TRUE FALSE
```

```r
summary(left_df)
```

```
  patient_id           site     visit_date         visit_year
 Length:240         Milan:96   Length:240         Length:240
 Class :character   Rome :77   Class :character   Class :character
 Mode  :character   Turin:67   Mode  :character   Mode  :character




 visit_month         visit_day              age           sex       smoking
 Length:240         Length:240         Min.   :26.0   Female:118   NA's:240
 Class :character   Class :character   1st Qu.:49.0   Male  :122
 Mode  :character   Mode  :character   Median :57.5
                                       Mean   :57.3
                                       3rd Qu.:65.0
                                       Max.   :85.0

   treatment  stage      weight_kg        height_cm     systolic_bp_mmhg
 Control:74   I  :76   Min.   : 39.40   Min.   :148.9   Min.   :115.0
 Drug_A :88   II :73   1st Qu.: 65.35   1st Qu.:162.2   1st Qu.:139.0
```

```
 Drug_B :78   III:53   Median : 75.65   Median :168.5   Median :146.5
              IV :38   Mean   : 76.87   Mean   :169.2   Mean   :146.7
                       3rd Qu.: 87.83   3rd Qu.:174.8   3rd Qu.:153.2
                       Max.   :132.90   Max.   :200.1   Max.   :192.0

 diastolic_bp_mmhg  glucose_mgdl   cholesterol_mgdl    crp_ngml
 Min.   : 73.00    Min.   : 60.0   Min.   :113.0    Min.   : 1.92
 1st Qu.: 90.75    1st Qu.: 98.0   1st Qu.:189.8    1st Qu.: 5.30
 Median : 97.00    Median :109.0   Median :207.0    Median : 8.37
 Mean   : 96.11    Mean   :108.8   Mean   :207.9    Mean   :10.35
 3rd Qu.:102.00    3rd Qu.:118.0   3rd Qu.:226.0    3rd Qu.:13.02
 Max.   :120.00    Max.   :149.0   Max.   :307.0    Max.   :42.09

    height_m          BMI              BMI_class   hemoglobin_gdl
 Min.   :1.489   Min.   :16.00   Underweight: 67   Min.   : 9.20
 1st Qu.:1.623   1st Qu.:23.77   Normal     : 58   1st Qu.:12.40
 Median :1.685   Median :26.85   Overweight :102   Median :13.25
 Mean   :1.692   Mean   :26.71   Obese      : 13   Mean   :13.50
 3rd Qu.:1.748   3rd Qu.:29.80                     3rd Qu.:14.70
 Max.   :2.001   Max.   :40.70                     Max.   :17.00
                                                   NA's   :80
   wbc_10e9l      platelets_10e9l     alt_ul       creatinine_mgdl
 Min.   : 2.310   Min.   : 60.0   Min.   : 10.00   Min.   :0.630
 1st Qu.: 5.518   1st Qu.:191.8   1st Qu.: 32.75   1st Qu.:1.018
 Median : 7.000   Median :232.0   Median : 46.50   Median :1.140
 Mean   : 6.758   Mean   :231.7   Mean   : 52.20   Mean   :1.138
 3rd Qu.: 7.902   3rd Qu.:271.0   3rd Qu.: 61.00   3rd Qu.:1.252
 Max.   :11.380   Max.   :391.0   Max.   :161.00   Max.   :1.580
 NA's   :80       NA's   :80      NA's   :80       NA's   :80
```

```r
# inner_join: keep only the 90 patients who have both clinical and lab data
matched_only <- clinical_data |>
  inner_join(lab_data, by = "patient_id")

nrow(matched_only)
```

```
[1] 160
```

In most clinical research scenarios, `left_join` is the most common choice: you start with your full patient list and enrich it with additional data, accepting that some patients may lack certain measurements (NAs).

**Exercise 2.4.**

1. Perform a `left_join` of `clinical_data` with `lab_data`.

2. Among patients who have lab data available, compute the mean hemoglobin and mean creatinine per treatment group.
3. Which group has the highest mean creatinine?

Solution

```
# 1. Left join
joined_df <- clinical_data |>
  left_join(lab_data, by = "patient_id")

# 2. Summarize (filter out NA first)
lab_summary <- joined_df |>
  filter(!is.na(hemoglobin_gdl)) |> # filter for what is NOT NA in hemoglobin
  group_by(treatment) |>
  summarize(mean_hemoglobin   = round(mean(hemoglobin_gdl), 2),
            mean_creatinine   = round(mean(creatinine_mgdl), 3),
            n                 = n(),
            .groups = "drop")

lab_summary
```

```
# A tibble: 3 x 4
  treatment mean_hemoglobin mean_creatinine     n
  <fct>               <dbl>           <dbl> <int>
1 Control              13.7            1.16    51
2 Drug_A               13.5            1.13    62
3 Drug_B               13.2            1.13    47
```

```
# 3. Highest mean creatinine
lab_summary |> arrange(desc(mean_creatinine))
```

```
# A tibble: 3 x 4
  treatment mean_hemoglobin mean_creatinine     n
  <fct>               <dbl>           <dbl> <int>
1 Control              13.7            1.16    51
2 Drug_B               13.2            1.13    47
3 Drug_A               13.5            1.13    62
```

---

## 2.4 Save your results

After all this work, let's save the enriched and cleaned dataset:

```r
# Save the left-joined dataset with all derived variables
final_data <- clinical_data |>
  left_join(lab_data, by = "patient_id")

write.csv(final_data,
          file = "output/clinical_data_enriched.csv",
          row.names = FALSE)
```

---

## 2.5 Home exercise

Try to do the following on your own:

1. Load `clinical_data_enriched.csv` from the output folder.
2. Clean up: keep only the columns `patient_id`, `site`, `sex`, `age`, `treatment`, `stage`, `BMI`, `BMI_class`, `glucose_mgdl`, `cholesterol_mgdl`, `crp_ngml`, `hemoglobin_gdl`, `creatinine_mgdl`.
3. Restrict to patients who have complete data (no NAs) — hint: use function `drop_na()`, see this.
4. For each stage, compute: number of patients, mean age, mean BMI, and proportion of obese patients.
5. Save the summary table to `output/summary_by_stage.csv`.

```r
library(tidyverse)

# 1. Load the enriched dataset
clinical_data_enriched <- read.csv("output/clinical_data_enriched.csv")

# 2. Keep only the columns of interest
clean_data <- clinical_data_enriched |>
  select(patient_id, site, sex, age, treatment, stage,
         BMI, BMI_class, glucose_mgdl, cholesterol_mgdl,
         crp_ngml, hemoglobin_gdl, creatinine_mgdl)

# 3. Restrict to patients with complete data (no NAs)
clean_data <- clean_data |>
  drop_na()

cat("Patients with complete data:", nrow(clean_data), "\n")
```

```
Patients with complete data: 160
```

```r
# 4. Summary by stage
summary_by_stage <- clean_data |>
  group_by(stage) |>
  summarize(
    n_patients       = n(),
    mean_age         = round(mean(age), 1),
    mean_BMI         = round(mean(BMI), 2),
    prop_obese       = round(mean(BMI_class == "Obese"), 3),
    .groups = "drop"
  )

print(summary_by_stage)
```

```
# A tibble: 4 x 5
  stage n_patients mean_age mean_BMI prop_obese
  <chr>      <int>    <dbl>    <dbl>      <dbl>
1 I             47     55.2     27.2      0.043
2 II            53     57.1     27.3      0.038
3 III           35     56.2     28.2      0.029
4 IV            25     57.5     24.2      0.16
```

```r
# 5. Save
write.csv(summary_by_stage,
          file = "output/summary_by_stage.csv",
          row.names = FALSE)
```

---

## 2.6 Bonus: how the clinical data were created

If you want to understand how we generated the dataset for this lesson, download the R script Notice the use of `set.seed` to make results reproducible.